VIVEK KUMAR SINGH

MSC DIGITAL ENGINEERING | DATA, AI AND AUTOMATION

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Work Authorisation: Student visa (Germany) - permitted to work part-time (Werkstudent, internship)

PROFESSIONAL SUMMARY

MSc Digital Engineering student with professional experience using Python, SQL, Pandas, PostgreSQL, Power BI, ETL workflows, REST APIs, and workflow automation. Built validated data pipelines, dashboards, and applied machine-learning applications across customer, supply-chain, mobility, clinical, and computer-vision use cases. Combines hands-on implementation with structured documentation and clear stakeholder communication.

EDUCATION

Master of Science, Digital Engineering | Otto-von-Guericke University, Magdeburg, Germany
Apr 2025 - Present | Current Grade: 1.7
Bachelor of Technology, Computer Science and Engineering | Abdul Kalam Technical University, India
Jul 2016 - Jun 2020 | Percentage: 80%

TECHNICAL SKILLS

Programming and Data: Python, SQL, Pandas, NumPy, JavaScript (Basic)
Data Engineering and Automation: ETL Pipelines, Data Preparation, Data Validation, REST API Integration, Workflow Automation
Machine Learning: Scikit-learn, TensorFlow, PyTorch, NLP, Predictive Modelling, YOLO11, SAM 2
Databases and Reporting: PostgreSQL, SQL Server, SQLAlchemy, Power BI, Excel, Matplotlib, Seaborn, PowerPoint
Platforms and Tools: Git, Docker, Streamlit, SAP HANA, SAP S/4HANA, SAP Datasphere, Salesforce, LeadSquared

WORK EXPERIENCE

Research Assistant | Otto-von-Guericke University, Magdeburg, Germany
Oct 2025 - Feb 2026
- Authored 10+ course modules covering data science, software development, API integration, and automation concepts.
- Designed practical use cases, UML and BPMN diagrams, and XML-based question sets for structured technical learning.
- Facilitated exercise classes for 30+ students and maintained 50+ learning assets on the university e-learning platform.

Data Analyst | Maxonic Inc., India
Oct 2022 - Mar 2025
- Cleaned and validated 3,900 customer-shopping records using Python and Pandas (standardised fields, removed duplicates).
- Created age-group and purchase-frequency features for customer segmentation and repeat-purchase analysis.
- Loaded the prepared data into PostgreSQL via SQLAlchemy and presented customer-behaviour findings in a Power BI dashboard.

Business Development Manager | BYJU'S, India
Jul 2020 - Sep 2022
- Analysed conversion trends in Salesforce and LeadSquared CRM and built Excel and Salesforce dashboards tracking 200+ prospects.
- Led and grew a sales team from 12 to 17 members, reaching peak weekly revenue of INR 2.1 million through structured targeting and performance tracking.

PROJECTS

E-Commerce Inventory Analysis: Evaluated 50K+ supply-chain records using Python, SQL, and machine learning for demand forecasting and logistics optimisation.
Clinical Risk Early Detection System: Applied NLP feature engineering with Logistic Regression and XGBoost; delivered an 85%-accuracy risk-reporting application through a Streamlit API.
Taxi Fleet Profit Optimization: Built a machine-learning fare-prediction and route-profitability system using NYC taxi trip data to support driver-earning and idle-time recommendations.
Conveyor Parcel Segmentation and Tracking: Applied YOLO11 and SAM 2 to detect and segment polybags on conveyor systems for logistics and parcel-handling applications.

CERTIFICATIONS

Developing AI Models with Machine Learning Clients for SAP HANA (SAP, 2024)
Setting up an ABAP Environment on SAP BTP (SAP, 2024)

LANGUAGES

English (C1) | German (A1, actively improving)